

Architecture-Aware Straight-Through Estimation for Robust Analog In-Memory Computing

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Abstract

Analog in-memory computing (IMC) can execute matrix–vector products with high parallelism, but programming error, drift, asymmetric nonlinearity, quantization, crosstalk, and read noise jointly make deployment operators both stochastic and non-smooth. We present an architecture-aware straight-through estimation framework that exposes one noisy operator across linear, dense convolution, grouped convolution, and depthwise convolution layers while using controlled surrogate gradients for training. The framework combines saturation-aware backward estimators with a forward error decomposition that separates stochastic read variation from systematic saturation bias. This decomposition motivates a clean-statistics activation preconditioner and layer-type-aware repeated reads without reducing the noise assigned to any individual operation. Under a strict protocol in which every convolution and linear layer uses the full challenge noise model, all six settings improve by 4.24–75.80 percentage points over direct noisy deployment, and four improve by at least 29.21 points. Five settings retain 86.60–94.78% of clean accuracy. On the most fragile case, TinyImageNet with EfficientNet-B0, direct deployment obtains 0.50% top-1 accuracy; preconditioned STE with depthwise and pointwise repeated reads reaches $76.30\% \pm 0.25\%$ (95% CI), versus 81.00% clean. A variance-guided 4–8-read policy reduces sensitive-layer reads by 25.5% and measured full-validation time by 23.0% with no significant accuracy change. We also identify the limits of backward-only variance scaling and report corrected shared-read semantics for memory-bounded large-image convolution.

1 Introduction

Analog IMC arrays reduce data movement by storing weights and performing multiply–accumulate operations in the same physical substrate [1–3]. Their analog nature, however, introduces errors that do not resemble a single independent Gaussian perturbation. Programmed conductances vary, drift and retention alter weights over time, nonlinear transfer functions saturate large partial sums, and readout circuits add quantization, spatially correlated noise, and global supply fluctuations. Hardware-aware training can recover substantial accuracy when the noise model is known [4], but a general training interface must also remain numerically useful when the simulated forward contains random sampling and non-differentiable quantization.

Straight-through estimation (STE) replaces an unusable local derivative with a surrogate during backpropagation [5]. It is widely used for binary and quantized networks [6–8], yet applying it to IMC raises two additional problems. First, the

forward error has both random and systematic components; repeated reads reduce the former but cannot remove saturation bias. Second, layer topology matters. A residual network dominated by dense 3×3 convolutions and an EfficientNet dominated by depthwise and pointwise operators can react very differently to the same nominal nonidealities.

This work studies that interaction under the 2026 INNO IMC challenge specification [9]. We deliberately use a strict evaluation protocol: every convolution and linear layer receives the complete noise configuration at scale 1.0. Layer-wise noise reduction is evaluated only as a diagnostic hardware-mapping bound and is excluded from the main results.

Our contributions are fourfold:

- We implement a common noisy/STE abstraction for linear, dense, grouped, and depthwise convolution, including memory-bounded convolution with physical read state shared across implementation chunks.
- We develop saturation-, variance-, and channel-aware surrogate gradients and derive a biased non-convex SGD bound that makes their intended bias–variance trade-off explicit.
- We show that systematic saturation, rather than read variance alone, dominates sensitive EfficientNet layers. A statistics-driven activation preconditioner preserves the clean linear map while keeping every physical noise coefficient unchanged.
- We evaluate six classification model–dataset pairs, perform repeated noisy evaluation and controlled ablations, quantify training cost on an RTX 4090, and report detection/segmentation extensions with explicit scope limitations.

2 Related Work

Analog IMC simulation and robust training. Resistive cross-point arrays have been studied for both dense and convolutional network training [1, 10] and for mixed-precision computation [2]. Hardware-aware noise injection can preserve near-digital inference accuracy on phase-change memory [4]. AI-HWKit provides configurable PyTorch simulation of crossbar nonidealities [11]. Noise-aware normalization [12] and error suppression/compensation [13] address activation distribution shift and error amplification. Our focus is complementary: a generic surrogate-gradient interface for a challenge-defined composite operator, with explicit treatment of architecture-dependent saturation and read semantics.

Surrogate gradients. The STE copies or modifies an upstream derivative through a stochastic or piecewise-constant operation [5]. BinaryConnect and binarized neural networks established its practical value [6, 7]; later analysis showed that a suitable coarse gradient can align with a population gradi-

ent in restricted quantized models [8]. Learned quantization methods also scale surrogate gradients to stabilize trainable ranges [14]. In contrast to quantization-only work, our forward includes stateful weight perturbations, asymmetric saturation, stochastic readout, and correlated/global effects. We therefore separate forward conditioning from backward confidence scaling and test whether each supplies independent empirical gain.

3 Problem Formulation

Let $X \in \mathbb{R}^{R \times I}$ and $W \in \mathbb{R}^{I \times O}$. A noisy IMC operator $\mathcal{N}_\phi(X, W; \xi)$ is parameterized by nonideality configuration ϕ and random state ξ . In compact form,

$$\tilde{Y} = [Q_b(h_{\alpha, \beta}((X + \Delta X)(W + \Delta W))) + \Delta Y] (1 + \delta_s), \quad (1)$$

where ΔW combines programming, drift, retention, and temperature terms; ΔX models input crosstalk; ΔY combines independent and spatially correlated output noise; and δ_s is a read-level supply variation. The asymmetric transfer function is

$$h_{\alpha, \beta}(z) = \begin{cases} \tanh(\alpha z)/\alpha, & z \geq 0, \\ \tanh((\alpha + \beta)z)/(\alpha + \beta), & z < 0. \end{cases} \quad (2)$$

The ADC operator Q_b rounds to a signed b -bit grid and adds sub-step quantization noise. Appendix A lists all coefficients.

For model parameters θ , input-label pair (x, y) , and task loss ℓ , the robust objective is

$$F(\theta) = \mathbb{E}_{(x, y)} \mathbb{E}_\xi [\ell(f_{\theta, \phi}(x; \xi), y)]. \quad (3)$$

Direct differentiation through Eq. (1) is ineffective at the quantizer and can have excessive variance elsewhere. We seek a surrogate $\hat{g}$ that remains informative while the forward pass exactly follows the specified noisy operator.

Strict-noise protocol. The main experiments set a global noise multiplier to one for all nine nonideality coefficients in every converted convolution and linear layer. We do not exempt first/last, depthwise, pointwise, or classifier layers. Repeated reads average independent applications of the same full-strength operator; they do not attenuate the noise of an individual read.

4 Architecture-Aware IMC-STE

4.1 Generic Layer Conversion

Model conversion recursively replaces Linear and Conv2d modules while copying their parameters. Convolution is expressed as an unfolded matrix product. Grouped convolution uses batched grouped products,

$$Y_{r, g, o} = \sum_{k=1}^{I_g} X_{r, g, k} W_{g, k, o}, \quad (4)$$

which keeps group weights isolated and makes depthwise convolution a special case. This gives all architectures the same

forward noise implementation and the same family of custom backward rules.

4.2 Surrogate-Gradient Family

For upstream gradient $G = \partial \ell / \partial \tilde{Y}$, identity STE uses the ideal local Jacobian:

$$\widehat{\nabla_X \ell} = G W^\top, \quad \widehat{\nabla_W \ell} = X^\top G. \quad (5)$$

Saturation-aware STE multiplies G by a softened derivative evaluated at the ideal preactivation $Z = XW$:

$$S(Z) = \gamma + (1 - \gamma) \begin{cases} 1 - \tanh^2(\alpha Z), & Z \geq 0, \\ 1 - \tanh^2((\alpha + \beta)Z), & Z < 0, \end{cases} \quad (6)$$

where γ prevents complete gradient suppression. Its adaptive variant normalizes S by the detached per-output mean and clips the result.

The variance-aware variant uses measured layer statistics. If $r_{v, l}$ and $r_{b, l}$ are stochastic- and bias-to-signal RMS ratios and K_l reads are averaged, layer confidence is

$$c_l = \max \left(c_{\min}, \left(1 + \lambda_v r_{v, l}^2 / K_l + \lambda_b r_{b, l}^2 \right)^{-1/2} \right). \quad (7)$$

An optional channel vector d_l further rescales G . Ordinary, depthwise, pointwise, and linear modules can carry separate profiles. This design encodes the fact that averaging reduces random variance approximately as $1/K_l$, but does not reduce systematic bias. Section 7 shows that this backward-only correction stabilizes gradients but is not the source of our largest accuracy improvement.

4.3 Statistics-Driven Activation Preconditioning

For each layer l , we collect the 99th percentile q_l of the absolute clean MAC output and define

$$a_l = \text{clip}(\tau / q_l, a_{\min}, 1), \quad \tilde{Y}_l = \frac{1}{a_l} \mathcal{N}_\phi(a_l X_l, W_l; \xi). \quad (8)$$

We use $\tau = 4$ and $a_{\min} = 0.1$ for depthwise and pointwise layers. In the clean linear limit, $a_l^{-1}(a_l X_l W_l) = X_l W_l$, so the intended operator is unchanged. Unlike shrinking W_l and amplifying the output, input scaling does not amplify the first-order programming perturbation $X_l \Delta W_l$. It does amplify additive output noise by $1/a_l$, which motivates the read allocation below. Statistics are frozen from the original clean checkpoint; re-estimating them after noisy fine-tuning was empirically harmful.

We also evaluate bounded learned scales,

$$a_l = a_{\min} + (a_{\max} - a_{\min}) \sigma(\vartheta_l), \quad R_a = \frac{1}{L_a} \sum_l (\log a_l - \log a_l^{(0)})^2, \quad (9)$$

initialized by Eq. (8). The constraint is essential: an unconstrained scale sweep drifts out of the validated linear range.

4.4 Read-Aware Training and Inference

Exact K_l -read inference returns

$$\bar{Y}_l = \frac{1}{K_l} \sum_{k=1}^{K_l} \mathcal{N}_\phi(X_l, W_l; \xi_{l,k}). \quad (10)$$

For EfficientNet-B0, training uses four reads on depth-wise/pointwise layers and evaluation uses eight. To compensate for output-noise amplification in Eq. (8), an adaptive policy assigns

$$K_l = \text{clip}(\lceil K_0/a_l^2 \rceil, K_0, K_{\max}). \quad (11)$$

We further implement a training-only moment-matched approximation that scales zero-mean random terms by $1/\sqrt{K_l}$ while retaining nonlinearity and ADC parameters. Final evaluation always uses exact reads.

4.5 Shared State for Chunked Convolution

Large images make full unfold tensors prohibitively large. We therefore chunk the output rows, but sample weight-side programming, drift, retention, temperature, and global supply state once per physical read and reuse it across all spatial chunks and MAC tiles. Input crosstalk and output read noise remain location-specific. Earlier per-chunk resampling introduced unintended spatial averaging, so chunk size changed not only memory but also the physical model. The shared-read state removes that confound while reducing the unfold workspace from $O(NH_o W_o C_i k_h k_w)$ to $O(Nh_c W_o C_i k_h k_w)$ for h_c output rows per chunk.

5 Optimization Analysis

Let $\hat{g}_t$ denote the STE gradient and write its conditional bias and variance as

$$\mathbb{E}_t[\hat{g}_t] = \nabla F(\theta_t) + b_t, \quad \mathbb{E}_t\|\hat{g}_t - \mathbb{E}_t\hat{g}_t\|^2 \leq \sigma_g^2. \quad (12)$$

The mean-squared estimation error is the sum of $\|b_t\|^2$ and the conditional variance. The online channel profile is a diagonal matrix D_t with entries in $[d_{\min}, 1]$, hence

$$d_{\min}\|g\|_2 \leq \|D_t g\|_2 \leq \|g\|_2. \quad (13)$$

It cannot amplify high-variance channels, while the floor prevents complete gradient loss.

Proposition 1. Assume F is L -smooth and lower bounded by F_* , $\|b_t\| \leq B$, and Eq. (12) holds. For $\theta_{t+1} = \theta_t - \eta \hat{g}_t$ with $0 < \eta \leq 1/(4L)$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla F(\theta_t)\|^2 \leq \frac{2(F(\theta_0) - F_*)}{\eta T} + 2(1 + L\eta)B^2 + L\eta\sigma_g^2. \quad (14)$$

The proof follows the standard smoothness descent argument [15] and is given in Appendix C. For an unbiased estimator and

$\eta = O(T^{-1/2})$, the average stationarity measure has the usual $O(T^{-1/2})$ behavior. Biased STE converges only to a neighborhood controlled by B^2 . This result motivates reducing systematic saturation bias and random read variance separately; it does not guarantee final classification accuracy, and the online profile’s dependence on the current batch requires additional assumptions for a sharper adaptive proof.

All clean, noisy, and single-read STE matrix products have leading arithmetic cost $\Theta(RIO)$. Exact K reads cost $\Theta(KRIO)$. Online paired profiling adds one clean forward to the noisy forward/backward but does not change the asymptotic order. Persistent profile storage is only $O(C_{out})$ per layer and is disabled at deployment.

6 Experiments

6.1 Setup

We evaluate ResNet18 [16] and EfficientNet-B0 [17] on CIFAR-10/100 [18] and TinyImageNet [19], an ImageNet-like 200-class subset derived from the ImageNet hierarchy [20]. TinyImageNet uses 224-pixel inputs and ImageNet initialization. CIFAR clean models are trained for 120 epochs with SGD, cosine decay, AutoAugment, and label smoothing; noisy fine-tuning starts from the corresponding clean checkpoint, uses gradient clipping at 1.0, and selects a checkpoint using noisy validation. TinyImageNet protocols are detailed in Appendix B.

We compare (i) clean digital evaluation, (ii) direct noisy deployment of the clean checkpoint, (iii) standard noisy training through the native simulated graph, and (iv) identity, saturation-aware, and adaptive saturation-aware STE. All reported main noisy results use the strict protocol in Section 3. CIFAR results are means over five independent noisy evaluations of a fixed trained checkpoint. TinyImageNet ResNet18 and direct EfficientNet results use three noisy evaluations; the final pre-conditioned EfficientNet result uses five. Thus these intervals measure deployment noise, not full retraining uncertainty. We report normal-approximation 95% confidence intervals and use paired t -tests when the same noise seeds are available; Welch tests are used for unpaired comparisons [21].

6.2 Strict Uniform-Noise Results

Table 1 reports the clean, direct noisy, and best strict STE result for each required classification pair. STE recovers more than 29 pp on both CIFAR-10 architectures and 33.84 pp on CIFAR-100 with EfficientNet-B0. CIFAR-100 with ResNet18 and TinyImageNet with ResNet18 are harder relative cases: their direct noisy checkpoints are already less damaged, and the best recovery is 6.98 pp and 4.24 pp , respectively.

Table 2 isolates the effect of the backward estimator on the four CIFAR settings. Native noisy training improves over direct deployment, but identity STE adds a further 6.71–15.96 pp . Saturation-aware scaling is most useful for CIFAR-10 EfficientNet-B0, while identity STE is marginally best on two settings. This variability argues for exposing a strategy

Table 1: Strict uniform-noise classification. Noisy entries are repeated-evaluation means; $\pm$ denotes 95% CI. Recovery is best STE minus direct noisy. No layer has reduced noise.

Dataset / architecture	Clean	Direct noisy	Best strict	Recovery	Retained	Best strict protocol
CIFAR-10 / ResNet18	95.62	61.42 $\pm$ 1.30	90.63 $\pm$ 0.19	+29.21	94.78	identity STE
CIFAR-10 / EfficientNet-B0	91.13	38.93 $\pm$ 0.60	86.12 $\pm$ 0.17	+47.19	94.50	saturation-aware STE
CIFAR-100 / ResNet18	78.32	62.98 $\pm$ 0.52	69.96 $\pm$ 0.33	+6.98	89.33	adaptive saturation STE
CIFAR-100 / EfficientNet-B0	71.14	27.76 $\pm$ 0.50	61.60 $\pm$ 0.32	+33.84	86.60	identity STE
TinyImageNet / ResNet18-224	71.60	53.83 $\pm$ 0.26	58.07 $\pm$ 0.53	+4.24	81.10	saturation-aware STE
TinyImageNet / EfficientNet-B0-224	81.00	0.50 $\pm$ 0.06	76.30 $\pm$ 0.25	+75.80	94.20	preconditioned STE + p/d read 8

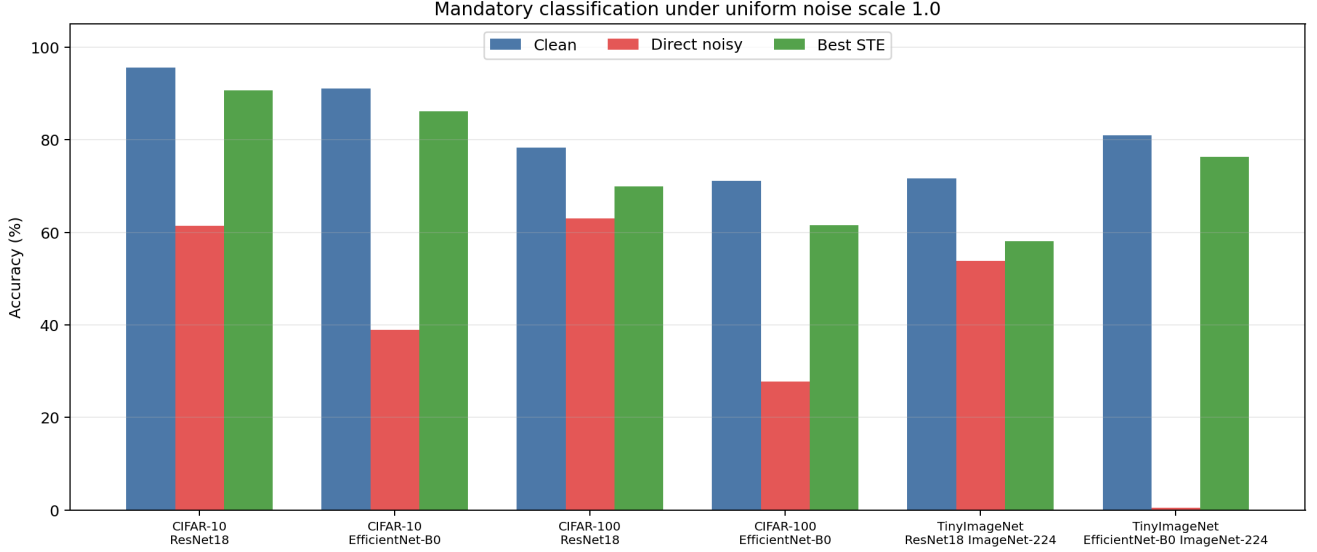


Figure 1: Clean, direct noisy, and best strict STE accuracy for all six classification pairs. EfficientNet-B0 on TinyImageNet is the clearest architecture-dependent failure: the clean checkpoint is nearly random under direct deployment, while the proposed preconditioning/read-aware pipeline restores most clean accuracy.

Table 2: Five-mode CIFAR comparison under the same strict noisy evaluation (top-1, %). “Native” differentiates the simulated graph directly.

Task	Direct	Native	STE	Sat.	Adapt.
C10 / R18	61.42	82.37	90.63	90.51	90.52
C10 / Eff.	38.93	68.44	84.40	86.12	86.03
C100 / R18	62.98	63.20	69.91	69.88	69.96
C100 / Eff.	27.76	48.92	61.60	61.49	61.36

family rather than assuming that a more elaborate surrogate is universally superior.

Table 3: TinyImageNet EfficientNet-B0 progression under strict per-read noise.

Protocol	Top-1 (%)
Clean digital	81.00
Direct noisy, one read	0.50 $\pm$ 0.06
Uniform STE fine-tune, one read	25.24 $\pm$ 0.32
Uniform STE, all-layer read 8	41.58
Read-adapted STE, all-layer read 8	42.61
P/D read-aware STE, P/D read 8	43.78
Preconditioned STE, P/D read 8	76.30 $\pm$ 0.25
Bounded learned scale, P/D read 8	76.37 $\pm$ 0.14
Adaptive 4–8 reads	76.20 $\pm$ 0.05

6.3 EfficientNet Requires Forward Conditioning

TinyImageNet EfficientNet-B0 exposes a failure that backward tuning alone does not solve. Plain uniform-noise STE fine-tuning reaches 25.24%; averaging eight reads at all layers reaches 41.58%, and restricting read-aware training and evaluation to pointwise/depthwise (P/D) layers reaches 43.78%. Activation preconditioning then raises the five-seed full-validation mean to 76.30%, a 32.52 *pp* improvement over the best read-only route.

To identify the cause, we run one clean and four indepen-

dent noisy forwards and decompose mean squared error into total, stochastic, and residual systematic components. For depthwise layers, total/stochastic/systematic RMS-to-signal ratios are 0.832/0.142/0.818; for pointwise layers they are 0.883/0.236/0.838. Thus most error energy is systematic. The large jump from 43.78% to 76.30% is consistent with moving pre-BN MAC values away from the asymmetric saturation regime, while repeated reads address the smaller random component.

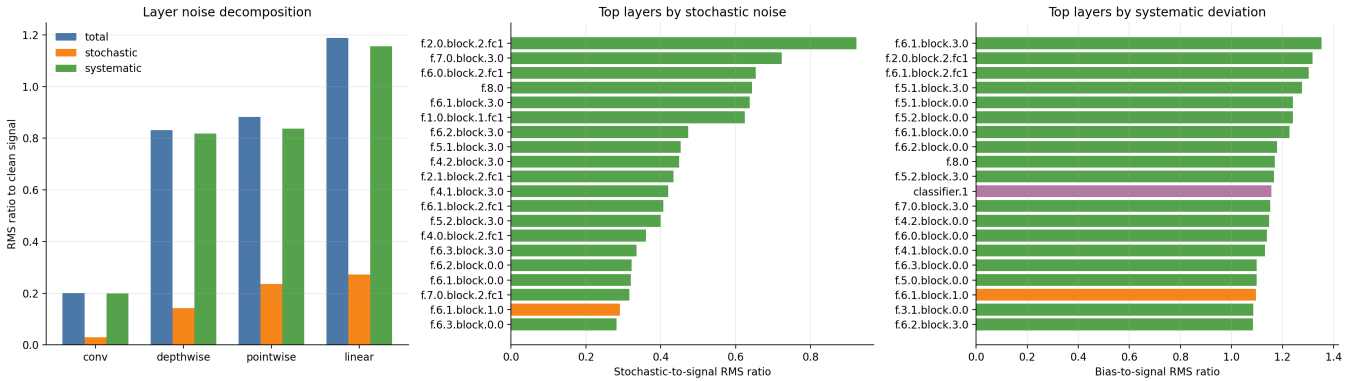


Figure 2: Layer error decomposition for the clean TinyImageNet EfficientNet-B0 checkpoint. Left: type averages. Center/right: layers with the largest stochastic and systematic components. Systematic deviation dominates depthwise and pointwise layers, motivating forward preconditioning rather than backward variance scaling alone.

7 Ablations and Efficiency

7.1 Estimator and Scale Ablations

On a matched 20-batch TinyImageNet comparison, backward-only variance/bias confidence obtains 41.95%, below the 42.27% read-aware control. With forward preconditioning, identity and variance-aware STE both obtain 74.45%, while the combined variance/saturation variant obtains 74.30%. The principal gain is therefore the forward transformation in Eq. (8), not a more complex backward rule. This negative result is important: the framework can express profile-aware gradients, but the data do not support claiming that they outperform identity STE in the main classification setting.

Bounded learned scales move the three-seed paired mean from 76.28% to 76.37%, an average $+0.09$ pp with $p = 0.247$. The effect is not significant. The adaptive read policy assigns 477 total reads across 80 sensitive layers versus 640 for uniform eight-read inference, a 25.47% reduction. Its mean accuracy is 0.073 pp lower with paired $p = 0.568$, also not significant. The same-seed full-validation runtime falls from 300.21 to 231.25 seconds (22.97%).

The training-only moment-matched approximation reduces a two-batch exact eight-read benchmark from 48.59 to 1.00 seconds per batch (48.7 $\times$), but joint fine-tuning with this approximation does not improve final accuracy. Accordingly, we use it only for low-cost training and retain exact reads for all formal evaluations.

7.2 Measured Training Cost

We benchmark a CIFAR-stem ResNet18 on an RTX 4090 with PyTorch 2.6.0 and CUDA 12.6, batch size 8, ten warm-up steps, and fifty measured full training steps. Each step includes forward, cross-entropy, backward, and SGD update. Online profiling includes its paired clean/noisy forwards.

Identity STE costs 5.28 $\times$ clean in this software simulator. Online profiling costs 1.87 $\times$ identity STE and adds 18.10 MiB incremental peak memory, mainly because of the extra clean forward. Profiling is training-only; its buffers are non-persistent

Table 4: RTX 4090 training-step microbenchmark. Peak memory is incremental allocated memory above the resident model baseline.

Path	Step (ms)	Images/s	Peak MiB
Clean	4.87	1643.94	35.50
Identity STE	25.71	311.12	278.35
Online profile STE	47.96	166.80	296.45

Table 5: Corrected shared-read full-validation stress test. STE uses one full fine-tuning epoch and strict uniform noise.

Task / metric	Clean	Direct	STE	Recovery
VOC07 Faster R-CNN / mAP50	72.59	28.72	37.53	+8.81
VOC12 DeepLabV3 / mIoU	71.31	4.03	6.46	+2.43

and incur no deployment inference cost. These numbers characterize this implementation, not analog hardware latency or energy.

8 Large-Image Cross-Domain Stress Test

We connect the same internal noisy layers to Faster R-CNN [22] and DeepLabV3 [23] on PASCAL VOC [24]. The purpose is to validate grouped conversion, memory-bounded unfold, shared-read state, and task-head diagnostics. These are not the challenge’s official COCO/Cityscapes optional benchmarks and must not be interpreted as completing that optional track.

The low absolute results are diagnostic, not hidden successes. Head-level residual-to-signal ratios exceed one for detection ROI and segmentation classifier modules. A combination of online channel profiling, pixel-logit consistency, and range control improves a matched segmentation evaluation by 0.245 pp over three training seeds ($p = 0.0086$, all seeds positive) and preserves +0.260 pp after one full epoch, but its 6.21% absolute mIoU does not exceed the 6.46% formal run. Detection profile and proposal-aligned ROI variants remain seed-unstable. We therefore retain the mechanism evidence without promoting these candidates to main results.

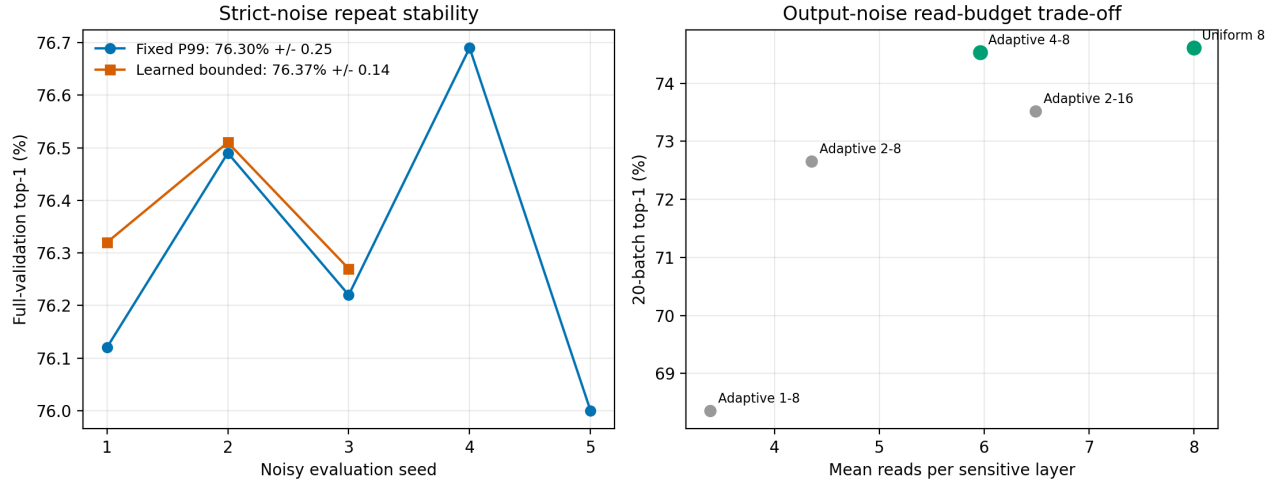


Figure 3: Left: repeated full-validation accuracy for fixed P99 and bounded learned activation scales. Learned-scale points are available for the first three matched seeds. Right: accuracy versus mean sensitive-layer reads for several allocation policies. Adaptive 4–8 reads preserve the uniform-eight operating point with lower logical read cost.

9 Limitations

First, the noise model is a software realization of the organizer-provided operator, not a calibration to a specific fabricated array. Accuracy cannot be compared directly with silicon measurements under different devices, mappings, or digital periphery. Second, most classification confidence intervals vary the noisy evaluation seed for one trained checkpoint; broader training-seed replication remains desirable. Third, TinyImageNet is an approved practical ImageNet-like task but is not ImageNet-1K. Fourth, multi-read inference trades latency and energy for variance reduction; we measure simulator runtime and logical reads, not hardware energy. Fifth, the VOC experiments validate the large-image mechanism but do not satisfy the official COCO/Cityscapes optional protocol, and their absolute noisy accuracy remains low. Finally, Eq. (14) is a generic biased-gradient bound under smoothness and bounded-moment assumptions, not a guarantee for arbitrary deep networks.

10 Conclusion

Complex IMC nonidealities require more than injecting scalar Gaussian noise. The proposed framework keeps the full noisy forward while providing generic surrogate gradients across dense, grouped, and depthwise operators. Across six strict classification settings, it recovers substantial deployment accuracy; on TinyImageNet EfficientNet-B0 it restores a nearly random 0.50% direct result to $76.30\% \pm 0.25\%$, retaining 94.20% of clean accuracy. Layer decomposition shows why: sensitive mobile convolutions are dominated by systematic saturation bias, so statistics-driven forward preconditioning is more effective than backward variance scaling alone. Read-aware training then addresses the remaining stochastic component, and an adaptive policy removes a quarter of sensitive-layer reads with-

out a significant accuracy change. The same implementation also exposes a crucial systems lesson for large images: memory chunking must not silently resample physical weight state. Together, these results support architecture-aware forward conditioning, explicit read semantics, and honest uncertainty reporting as central ingredients of robust IMC training.

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A Noise Configuration

Table 6 lists the full-strength configuration used in all strict experiments. The global noise multiplier scales every floating-point nonideality coefficient together and equals one in the main results. It does not alter the eight-bit ADC setting.

Table 6: Strict noise parameters.

Parameter	Value	Parameter	Value
Programming std.	0.010	Output std.	0.010
Drift factor	0.005	Crosstalk factor	0.002
Nonlinear α	0.100	Temperature factor	0.001
Asymmetry β	0.050	Retention loss	0.001
Supply std.	0.010	ADC bits	8

Programming noise is additive. Drift is proportional to $|W|$, retention is proportional to W , and temperature noise is proportional to $\sqrt{|W|}$. Input crosstalk scales with each input row’s vector norm. Output noise contains an independent component and a 3×3 spatially averaged component with 0.3 relative amplitude. Supply variation is scalar within a physical read.

B Protocol Details

CIFAR. Clean pretraining uses 120 full epochs. The formal five-mode comparison uses a two-stage clean-to-noisy procedure, learning rate 0.001 for the final 20-epoch sweep, and gradient clipping at 1.0. Earlier 40-epoch calibration runs are retained for saturation-aware CIFAR-10 checkpoints when they provide the best repeated noisy mean. Comparisons within each task use the same strict noisy evaluation configuration and full test set.

TinyImageNet ResNet18. The 224-input model uses ImageNet initialization. The best strict result uses saturation-aware STE at learning rate 5×10^{-4} for ten epochs with 300 training batches per epoch. Full validation contains 10,000 samples.

TinyImageNet EfficientNet-B0. The preconditioned route starts from the 81.00% clean checkpoint. Stage one uses identity STE, four depthwise/pointwise reads, 50 batches, and learning rate 5×10^{-4} . Stage two uses three epochs of 100 batches at 2×10^{-4} . Deployment evaluation uses eight exact reads on depthwise and pointwise layers and one read elsewhere. P99 statistics come from the original clean checkpoint, target $\tau = 4$, and floor $a_{\min} = 0.1$.

Checkpoint and uncertainty scope. Noisy evaluation repetitions vary simulator random seeds while holding the selected checkpoint fixed. Main classification CIs therefore quantify stochastic deployment variation conditional on training. Learned-scale and read-budget comparisons use matched noisy seeds, while the optional online-profile extension uses independently trained paired seeds.

C Proof of Proposition 1

By L -smoothness and the update rule,

$$\mathbb{E}_t F(\theta_{t+1}) \leq F(\theta_t) - \eta \langle \nabla F, \nabla F + b_t \rangle + \frac{L\eta^2}{2} \mathbb{E}_t \|\widehat{g}_t\|^2. \quad (15)$$

Young’s inequality gives $\langle \nabla F, b_t \rangle \geq -\frac{1}{4}\|\nabla F\|^2 - \|b_t\|^2$, and

$$\mathbb{E}_t \|\widehat{g}_t\|^2 \leq 2\|\nabla F\|^2 + 2B^2 + \sigma_g^2. \quad (16)$$

Substitution and $L\eta \leq 1/4$ imply

$$\frac{\eta}{2} \mathbb{E} \|\nabla F(\theta_t)\|^2 \leq \mathbb{E}[F(\theta_t) - F(\theta_{t+1})] \quad (17)$$

$$+ \eta(1 + L\eta)B^2 + \frac{L\eta^2}{2} \sigma_g^2. \quad (18)$$

Summing from $t = 0$ to $T - 1$, using $F(\theta_T) \geq F_*$, and dividing by $\eta T/2$ yields Eq. (14).

D Additional Diagnostics

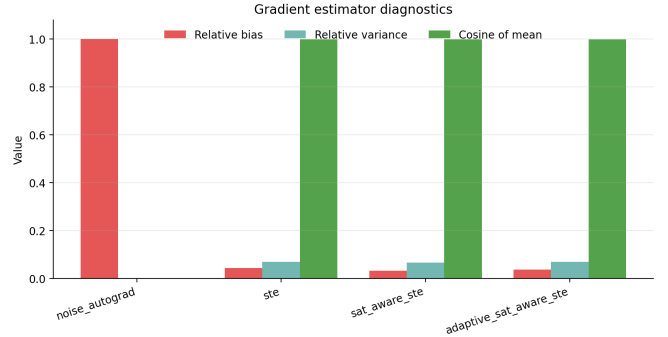


Figure 4: Gradient diagnostic comparing native noisy autograd, identity STE, and saturation-aware estimators. Native differentiation has a zero weight gradient through the quantized path in this probe; saturation-aware STE has mean-gradient cosine 0.9995 and relative bias 0.0331.

Layer-wise calibrated TinyImageNet EfficientNet-B0 reaches $80.09\% \pm 0.12\%$ by assigning lower noise coefficients to selected pointwise and linear layers. This value is useful as a hardware-mapping upper bound, but it violates the strict uniform-noise protocol and is excluded from all main rankings. Similarly, bypassing depthwise layers is retained only as a diagnostic that identifies topology sensitivity.

E Extended-Task Diagnostics and Scope

The reported large-image results use physical shared-read state: weight-side programming, drift, retention, temperature, and supply draws are reused across all implementation chunks belonging to one physical read. Input crosstalk and output noise remain location-specific. This makes the result invariant to the chosen memory-bounding chunk size.

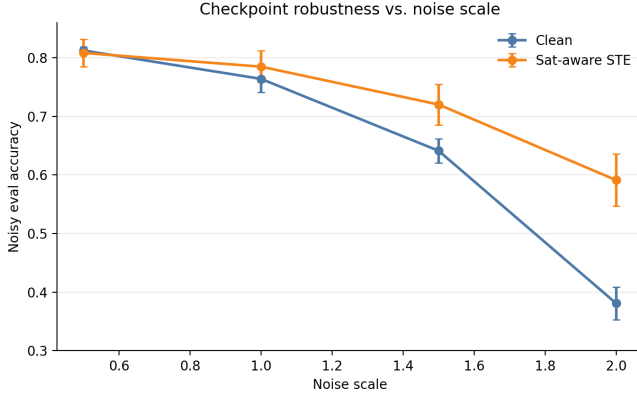


Figure 5: Checkpoint robustness as the complete noise configuration is scaled. STE-trained checkpoints degrade more slowly than direct clean-to-noisy deployment at high noise. This sweep is diagnostic; the main tables always use scale 1.0.

Corrected subsystem diagnostics give residual-to-signal ratios of 0.523, 0.676, 0.825, and 1.096 for the detection backbone, FPN, RPN, and ROI modules. The segmentation backbone, classifier, and auxiliary classifier obtain 0.756, 1.077, and 0.913. These finite, shape-consistent traces indicate a genuine head-sensitivity problem rather than an execution failure.

Three-seed paired comparisons require a positive mean delta, all seeds positive, and paired $p < 0.05$ for statistical support. Online segmentation satisfies these criteria with +0.245 pp , but does not improve the best absolute full-validation result after one epoch. Online detection and proposal-aligned ROI supervision retain at least one negative seed and are reported as non-significant mechanism studies.

F Reproducibility

The implementation uses PyTorch [25]. The repository contains the environment specification, fixed experiment configurations, dataset-free smoke test, 48 unit tests, table reconstruction, and artifact hash verification. From the repository root, the compact integrity check is:

```
conda env create -f environment.yml
conda activate imc-ste
python scripts/verify_project.py
```

The verification command runs the clean/noisy/STE and online-profile smoke paths, the test suite, arithmetic checks for reported results, and deterministic regeneration of the key table and figure. Public datasets and large checkpoints are intentionally excluded from the compact submission archive; the repository README records data preparation, full training, and checkpoint evaluation commands.